

Second ML FE Handoff + Response Live

Ringkasan

- `modelKey = heart_disease_v1`
 - `mlVersion` configurable via env
 - default live saat ini: `heart-risk-v1`
 - FE tetap hit backend Express, **bukan** service Python langsung
-

Kontrak assessment final: 9 fitur

`POST /users/{userId}/heart-risk-model/assessments`

`PUT /users/{userId}/heart-risk-model/assessments/{assessmentId}`

Body:

```
{
  "assessmentDate": "2026-06-02",
  "age": 58,
  "sex": 0,
  "chest_pain_type": 3,
  "resting_bp_s": 151,
  "fasting_blood_sugar": 0,
  "max_heart_rate": 118,
  "exercise_angina": 0,
  "old_peak": 0,
  "st_slope": 2
}
```

Patient endpoints

1. GET /users/{userId}/heart-risk-model/readiness

Query params: tidak ada

Response live

```
{
  "success": true,
  "message": "Status readiness second ML pasien berhasil diambil",
  "data": {
    "ready": true,
    "modelKey": "heart_disease_v1",
    "missingFields": [],
    "resolvedFields": [
      "age",
      "sex",
      "chest_pain_type",
      "resting_bp_s",
      "fasting_blood_sugar",
      "max_heart_rate",
      "exercise_angina",
      "old_peak",
      "st_slope"
    ],
    "sourceSummary": {
      "assessmentId": "03677278-9acf-4aeb-9f7b-f953b5120162",
      "assessmentDate": "2026-06-03",
      "derived": {
        "ageFromProfile": false,
        "sexFromProfile": false,
        "restingBpFromBodyMetric": false,
        "maxHeartRateFromBodyMetric": false
      },
      "latestBodyMetricMeasuredAt": null
    }
  }
}
```

2. GET /users/{userId}/heart-risk-model/assessment/latest

Query params: tidak ada

Response live

```
{
  "success": true,
  "message": "Assessment second ML terbaru pasien berhasil diambil",
  "data": {
    "assessmentId": "03677278-9acf-4aeb-9f7b-f953b5120162",
    "patientId": "29a8cae4-ba34-4418-baf0-0ba776fbec6b",
    "assessmentDate": "2026-06-03",
    "age": 58,
    "sex": 0,
    "chest_pain_type": 3,
    "resting_bp_s": 151,
    "fasting_blood_sugar": 0,
    "max_heart_rate": 118,
    "exercise_angina": 0,
    "old_peak": 0,
    "st_slope": 2,
    "createdAt": "2026-06-02T09:30:20.229Z",
    "updatedAt": "2026-06-02T09:30:20.553Z"
  }
}
```

3. GET /users/{userId}/heart-risk-model/assessments

Query params opsional:

- `page`
- `limit`

Catatan:

- ini buat list assessment
- shape item mirip `assessment/latest` , plus pagination

4. POST /users/{userId}/heart-risk-model/predictions

Query params opsional:

- `includePayload=false|true`

Body:

```
{}
```

Response live

```
{
  "success": true,
  "message": "Prediksi second ML pasien berhasil diambil",
  "data": {
    "resultId": "91b6c736-18c4-46af-9f0c-3d9d429ae8ee",
    "modelKey": "heart_disease_v1",
    "mlVersion": "heart-risk-v1",
    "generatedAt": "2026-06-02T09:29:52.943Z",
    "sourceSummary": {
      "assessmentId": "c6aef02e-96a3-4101-9118-eeb039041966",
      "assessmentDate": "2026-06-02"
    },
  },
  "upstream": {
    "status": 200,
    "body": {
      "modelKey": "heart_disease_v1",
      "mlVersion": "heart-risk-v1",
      "riskLevel": "low",
      "threshold": 0.43,
      "probability": 0.3695482015609741,
      "featureOrder": [
        "age",
        "sex",
        "chest_pain_type",
        "resting_bp_s",
        "fasting_blood_sugar",
        "max_heart_rate",
        "exercise_angina",
        "old_peak",
        "st_slope"
      ],
      "predictedClass": 0
    }
  }
}
```

```
}  
}
```

5. GET /users/{userId}/heart-risk-model/predictions/latest

Query params: tidak ada

Response live

```
{  
  "success": true,  
  "message": "Prediksi second ML terbaru pasien berhasil diambil",  
  "data": {  
    "resultId": "91b6c736-18c4-46af-9f0c-3d9d429ae8ee",  
    "patientId": "29a8cae4-ba34-4418-baf0-0ba776fbec6b",  
    "requestedByUserId": "29a8cae4-ba34-4418-baf0-0ba776fbec6b",  
    "modelKey": "heart_disease_v1",  
    "inferenceType": "prediction",  
    "requestContext": "patient",  
    "mlVersion": "heart-risk-v1",  
    "payloadHash": "sha256:0ea26adef035e4468951f2672823e635149bd4fc98c97c",  
    "payload": null,  
    "sourceSummary": {  
      "derived": {  
        "ageFromProfile": false,  
        "sexFromProfile": false,  
        "restingBpFromBodyMetric": false,  
        "maxHeartRateFromBodyMetric": false  
      },  
      "assessmentId": "c6aef02e-96a3-4101-9118-eeb039041966",  
      "assessmentDate": "2026-06-02",  
      "latestBodyMetricMeasuredAt": null  
    },  
    "window": {  
      "startDate": "2026-06-02",  
      "endDate": "2026-06-02"  
    },  
    "upstream": {  
      "endpoint": "http://heart-risk-ml-service:8090/v1/predictions",  
      "status": 200,  
    }  
  }  
}
```

```
    "body": {
      "modelKey": "heart_disease_v1",
      "mlVersion": "heart-risk-v1",
      "riskLevel": "low",
      "threshold": 0.43,
      "probability": 0.3695482015609741,
      "featureOrder": [
        "age",
        "sex",
        "chest_pain_type",
        "resting_bp_s",
        "fasting_blood_sugar",
        "max_heart_rate",
        "exercise_angina",
        "old_peak",
        "st_slope"
      ],
      "predictedClass": 0
    }
  },
  "generatedAt": "2026-06-02T09:29:52.943Z",
  "createdAt": "2026-06-02T09:29:52.945Z"
}
```

6. GET /users/{userId}/heart-risk-model/predictions/history

Query params optional:

- `page`
- `limit`

Response live

```
{
  "success": true,
  "message": "Riwayat prediksi second ML pasien berhasil diambil",
  "data": {
    "items": [
      {
```

```
    "resultId": "91b6c736-18c4-46af-9f0c-3d9d429ae8ee",
    "patientId": "29a8cae4-ba34-4418-baf0-0ba776fbec6b",
    "requestedByUserId": "29a8cae4-ba34-4418-baf0-0ba776fbec6b",
    "modelKey": "heart_disease_v1",
    "inferenceType": "prediction",
    "requestContext": "patient",
    "mlVersion": "heart-risk-v1",
    "window": {
      "startDate": "2026-06-02",
      "endDate": "2026-06-02"
    },
    "generatedAt": "2026-06-02T09:29:52.943Z",
    "createdAt": "2026-06-02T09:29:52.945Z"
  }
],
"pagination": {
  "page": 1,
  "limit": 5,
  "totalItems": 1,
  "totalPages": 1
}
}
```

7. GET /users/{userId}/heart-risk-model/predictions/history/{resultId}

Query params: tidak ada

Response live

```
{
  "success": true,
  "message": "Detail riwayat prediksi second ML pasien berhasil diambil",
  "data": {
    "resultId": "91b6c736-18c4-46af-9f0c-3d9d429ae8ee",
    "patientId": "29a8cae4-ba34-4418-baf0-0ba776fbec6b",
    "requestedByUserId": "29a8cae4-ba34-4418-baf0-0ba776fbec6b",
    "modelKey": "heart_disease_v1",
    "inferenceType": "prediction",
    "requestContext": "patient",
```

```
"mlVersion": "heart-risk-v1",
"payloadHash": "sha256:0ea26adef035e4468951f2672823e635149bd4fc98c97c",
"payload": null,
"sourceSummary": {
  "derived": {
    "ageFromProfile": false,
    "sexFromProfile": false,
    "restingBpFromBodyMetric": false,
    "maxHeartRateFromBodyMetric": false
  },
  "assessmentId": "c6aef02e-96a3-4101-9118-eeb039041966",
  "assessmentDate": "2026-06-02",
  "latestBodyMetricMeasuredAt": null
},
"window": {
  "startDate": "2026-06-02",
  "endDate": "2026-06-02"
},
"upstream": {
  "endpoint": "http://heart-risk-ml-service:8090/v1/predictions",
  "status": 200,
  "body": {
    "modelKey": "heart_disease_v1",
    "mlVersion": "heart-risk-v1",
    "riskLevel": "low",
    "threshold": 0.43,
    "probability": 0.3695482015609741,
    "featureOrder": [
      "age",
      "sex",
      "chest_pain_type",
      "resting_bp_s",
      "fasting_blood_sugar",
      "max_heart_rate",
      "exercise_angina",
      "old_peak",
      "st_slope"
    ],
    "predictedClass": 0
  }
},
"generatedAt": "2026-06-02T09:29:52.943Z",
"createdAt": "2026-06-02T09:29:52.945Z"
```



```
}
```

```
}
```

Doctor dashboard endpoints

1. GET

`/doctors/{doctorId}/dashboard/patients/{patientId}/heart-risk-model/readiness`

Response live

```
{
  "success": true,
  "message": "Status readiness second ML pasien dashboard berhasil diambi",
  "data": {
    "ready": true,
    "modelKey": "heart_disease_v1",
    "missingFields": [],
    "resolvedFields": [
      "age",
      "sex",
      "chest_pain_type",
      "resting_bp_s",
      "fasting_blood_sugar",
      "max_heart_rate",
      "exercise_angina",
      "old_peak",
      "st_slope"
    ],
    "sourceSummary": {
      "assessmentId": "03677278-9acf-4aeb-9f7b-f953b5120162",
      "assessmentDate": "2026-06-03",
      "derived": {
        "ageFromProfile": false,
        "sexFromProfile": false,
        "restingBpFromBodyMetric": false,
        "maxHeartRateFromBodyMetric": false
      },
      "latestBodyMetricMeasuredAt": null
    }
  }
}
```

```
}  
}
```

2. GET

`/doctors/{doctorId}/dashboard/patients/{patientId}/heart-risk-model/assessment/latest`

Response live

```
{  
  "success": true,  
  "message": "Assessment second ML terbaru pasien dashboard berhasil dian  
  "data": {  
    "assessmentId": "03677278-9acf-4aeb-9f7b-f953b5120162",  
    "patientId": "29a8cae4-ba34-4418-baf0-0ba776fbec6b",  
    "assessmentDate": "2026-06-03",  
    "age": 58,  
    "sex": 0,  
    "chest_pain_type": 3,  
    "resting_bp_s": 151,  
    "fasting_blood_sugar": 0,  
    "max_heart_rate": 118,  
    "exercise_angina": 0,  
    "old_peak": 0,  
    "st_slope": 2,  
    "createdAt": "2026-06-02T09:30:20.229Z",  
    "updatedAt": "2026-06-02T09:30:20.553Z"  
  }  
}
```

3. POST

`/doctors/{doctorId}/dashboard/patients/{patientId}/heart-risk-model/predictions`

Query params opsional:

- `includePayload=false|true`

Body:

```
{}
```

Catatan:

- shape response akan sama seperti prediction patient, tapi `requestContext` nanti `doctor` kalau dipicu dari sisi dokter
-

4. GET

`/doctors/{doctorId}/dashboard/patients/{patientId}/heart-risk-model/predictions/latest`

Response live

```
{
  "success": true,
  "message": "Prediksi second ML terbaru pasien dashboard berhasil diambil",
  "data": {
    "resultId": "91b6c736-18c4-46af-9f0c-3d9d429ae8ee",
    "patientId": "29a8cae4-ba34-4418-baf0-0ba776fbec6b",
    "requestedByUserId": "29a8cae4-ba34-4418-baf0-0ba776fbec6b",
    "modelKey": "heart_disease_v1",
    "inferenceType": "prediction",
    "requestContext": "patient",
    "mlVersion": "heart-risk-v1",
    "payloadHash": "sha256:0ea26adef035e4468951f2672823e635149bd4fc98c97c",
    "payload": null,
    "sourceSummary": {
      "derived": {
        "ageFromProfile": false,
        "sexFromProfile": false,
        "restingBpFromBodyMetric": false,
        "maxHeartRateFromBodyMetric": false
      },
      "assessmentId": "c6aef02e-96a3-4101-9118-eeb039041966",
      "assessmentDate": "2026-06-02",
      "latestBodyMetricMeasuredAt": null
    },
    "window": {
```

```
    "startDate": "2026-06-02",
    "endDate": "2026-06-02"
  },
  "upstream": {
    "endpoint": "http://heart-risk-ml-service:8090/v1/predictions",
    "status": 200,
    "body": {
      "modelKey": "heart_disease_v1",
      "mlVersion": "heart-risk-v1",
      "riskLevel": "low",
      "threshold": 0.43,
      "probability": 0.3695482015609741,
      "featureOrder": [
        "age",
        "sex",
        "chest_pain_type",
        "resting_bp_s",
        "fasting_blood_sugar",
        "max_heart_rate",
        "exercise_angina",
        "old_peak",
        "st_slope"
      ],
      "predictedClass": 0
    }
  },
  "generatedAt": "2026-06-02T09:29:52.943Z",
  "createdAt": "2026-06-02T09:29:52.945Z"
}
```

5. GET

/doctors/{doctorId}/dashboard/patients/{patientId}/heart-risk-model/predictions/history

Query params opsional:

- `page`
- `limit`

Catatan:

- shape item sama seperti history patient
-

6. GET

`/doctors/{doctorId}/dashboard/patients/{patientId}/heart-risk-model/predictions/history/{resultId}`

Catatan:

- shape detail sama seperti detail history patient
-

Error case live yang penting

Kalau assessment belum lengkap, readiness akan `false` dan prediction ditolak.

Readiness live saat incomplete

```
{
  "success": true,
  "data": {
    "ready": false,
    "modelKey": "heart_disease_v1",
    "missingFields": [
      "exercise_angina",
      "old_peak",
      "st_slope"
    ],
    "resolvedFields": [
      "age",
      "sex",
      "chest_pain_type",
      "resting_bp_s",
      "fasting_blood_sugar",
      "max_heart_rate"
    ]
  }
}
```

Prediction status saat incomplete

Catatan FE penting

- `predictedClass = 0` itu valid, bukan error
- angka utama untuk UI adalah `probability`
- `riskLevel` sekarang datang dari backend
- `mlVersion` sekarang configurable via env, tapi default live tetap:

```
"mlVersion": "heart-risk-v1"
```

- FE jangan hardcode `mlVersion` , cukup tampilkan dari backend kalau perlu
- flow yang benar:
 1. save assessment
 2. hit readiness
 3. kalau `ready=true` , baru hit prediction
 4. ambil latest/history/detail dari backend